ADVANCED PROBLEMS FOR LECTURE 14

In the lecture, we will be talking at a more general level of the patterns of what is going on. But there are deeper questions to explore therein, which we cannot get to. This is your opportunity to investigate deeper for yourself. Don't limit yourself to just these questions, let your curiosity guide you too! That is, if you wonder about any details, or if any certain patterns or ideas exist, see if you can state your question clearly, and answer it! Here are a few suggestions to get you started.

This problem set introduces the notion of topology and deals with some topology on $\mathbb{R}$. You should only need to know basic operations with sets to be able to do this.

Let's start with a couple of definitions.

Definition 0.1. A subset $S \subseteq \mathbb{R}$ is called *open* if it can be written as a (possibly empty) union of open intervals. We say that $T \subseteq \mathbb{R}$ is *closed* if $\mathbb{R} \setminus T$ is open.

(1) Show that closed intervals are closed. Show that $\mathbb{R}$ is closed. Show that $\emptyset$ is closed.
(2) Give an example of a closed set which is not a closed interval.
(3) Show that an intersection of two open sets is open. But that an infinite intersection need not be open.
(4) Show that any union of open sets is open.
(5) Give an example of an infinite closed set C which has no interior (i.e. there is no non-empty open set S for which $S \subseteq C$).
(6) Challenge: Show that a function $f : \mathbb{R} \to \mathbb{R}$ is continuous if and only if for every open interval I, the pre-image $f^{-1}(I) := \{x \in \mathbb{R} : f(x) \in I\}$ is an open set.

(7) It is nice to know when optimization problems have solutions. Part of what we can build up with topology here is a slightly better version of the Extreme Value Theorem. In general, the EVT on $\mathbb{R}$ says that the continuous image of a closed and bounded set is itself closed and bounded. However, this might be too complicated to prove right now. Instead, show that the continuous image of a closed set need not be closed, and show that the continuous image of a bounded set need not be bounded.

(8) Prove that L'Hôpital's Rule works. (Hint: Use a result from the last problem set)

"Topology" is a notion of closeness or connectedness of space. Formally speaking, a "topology" is just a set with a set of open sets on it. The rigorous definition is as follows:

Definition 0.2. A *topological space* is a pair $(\mathcal{X}, \mathcal{T})$ where $\mathcal{X}$ is a set, and $\mathcal{T}$ is a collection of subsets of $\mathcal{X}$ with the properties that

- $\emptyset \in \mathcal{T}$ and $\mathcal{X} \in \mathcal{T}$
- If $\{A_i\}_{i \in I} \subseteq \mathcal{T}$ then $\bigcup_{i \in I} A_i \in \mathcal{T}$. That is, $\mathcal{T}$ is closed under arbitrary unions of its elements.
- If $A_1, \ldots, A_n \in \mathcal{T}$ then $A_1 \cap \cdots \cap A_n \in \mathcal{T}$. That is, $\mathcal{T}$ is closed under finite intersections.

We call the elements of $\mathcal{T}$, *open sets*. You can check for yourself that $(\mathbb{R}, \mathcal{T})$ is a topological space, where $\mathcal{T}$ is the set of open sets that we have defined at the start.